

ABOUT

Senior AI Engineer at the intersection of computer science and finance. I ramp up quickly, listen for what clients actually need, and work full stack, from architecture and agents through UI and shipping. I move fast, stay close to the people who use the product, and keep systems trustworthy in production. At a large Canadian pension fund I owned an agentic AI platform through the full cycle, from proof of concept to deployment to years of production use with active users across the organization. Outside that corporate work I maintain digithings, an open-source AI infrastructure stack, and build tools on top of it for private clients.

PROJECTS

digithings.ai

Open-source AI library and toolkit. It is a canvas for agentic ideas and the layer I use to build production applications. Composable modules for orchestration, RAG/retrieval, DigiChat surfaces, auth, and observability; self-hosted by default (MIT), BYOK, swappable vector backends.

- DataTap Stream Chat:** DigiChat release embedded on a client’s product documentation, with support chat and developer/docs chat in place and retrieval against the live docs.

digiquant.io

Automated quant finance application built on digithings. Agentic research produces ranked, sourced theses; multi-agent deliberation shapes portfolio direction and risk sizing; then indicators, strategy specs, Optuna search, and reproducible NautilusTrader backtests with trade ledgers and tearsheets. Research book and paper portfolio ship today; live broker execution is not built.

- FX Hub:** FX research with traders on bank and broker consensus and trade ideas.

EXPERIENCE

Senior AI Engineer

Alpha Science · PSP Investments · Montreal, QC Sep 2022 – Aug 2026

- Owned the full stack of PSP Investments’ company-wide agentic AI platform for analysts and PMs: architecture, design documentation, and UI/UX from concept to deployment; feature development, SDK packaging & distribution, PR reviews, and ongoing maintenance. Source-traced answers land in daily research workflows.
- Alpha Science SDK:** wraps Databricks (Vector Search, SQL Warehouses, Genie Agents) & OpenAI (Responses + Files API), auto-auth via Databricks CLI & Azure Key Vault. Composable agent/workflow base-class framework on the firm’s existing stack, proprietary tracing, inter-agent persistence, pre-built financial utility agents & skills; artifact export (Office docs & web pages), sub-agent delegation with to-do planning, and parallel tool execution.
- Virtual Analyst Platform:** custom React, NestJS, and FastAPI frontend (refactored from Gradio) that became the shell for company AI tools and dashboards. Modern chat UI with a side panel for source citation and artifact viewing; workflow launcher for long-running research tasks; task manager with subscription-style pre-written prompts (e.g. digests). Step-level tracing, source traceability, structured data analysis, and live charting (Plotly/Dash/ECharts).
- Document rebuild pipeline:** from messy retrieval-index text, parallel LLM reformatting into Markdown/PDF reconstructions of filings, transcripts, and broker research so users can read full sourced documents in context, with the original text kept as source of truth.
- Agile roadmap:** weekly-evolving priorities with rapid integration of emerging model and tooling capabilities into production. Each feature independently scoped, designed as an SDK tool, and surfaced through the platform from concept to deployment.
- Leadership:** direct collaboration with C-suite across all asset classes; mentored interns and juniors to ship production features independently; coached Alpha Science SDK users; platform demoed at the Databricks Data & AI Summit.

React NestJS FastAPI Agentic AI RAG Observability

Analyst

Digital Innovation · PSP Investments · Montreal, QC Sep 2021 – Aug 2022

- Co-developed **Earnings Prediction Engine** (ML beat/miss pipeline, point-in-time data, Power BI); built alt-data pipeline with data profiling, cleaning & feature engineering for equity factor research; delivered POC analyses to drive stakeholder buy-in across asset class teams

Quant Strategy Alt Data ML Pipelines Power BI scikit-learn

Intern

Information Security · PSP Investments · Montreal, QC May 2021 – Aug 2021

- Enterprise risk framework implementation; automated risk assessment repository updates, improving data currency and reporting efficiency

Cybersecurity Risk Analysis

Business Analyst

Lunch à Porter · Montreal, QC Jun 2020 – Jun 2021

- SEO & A/B testing drove 2x online conversion rate; inventory analysis for data-informed merchandising decisions

SEO A/B Testing Shopify Lightspeed Analytics

CONTACT

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SKILLS

GEN AI & LLMS

LangGraph LangChain LiteLLM

RAG OpenAI API Gemini API

Claude API MCP Gradio

Azure AI Search Azure AI Foundry

Prompt Engineering Vector DBs

AI DEV TOOLS

Cursor GitHub Copilot

Claude Code Codex OpenCode

LM Studio Open WebUI Ollama

DATA & CLOUD

Azure GCP Databricks Spark

Delta Lake PostgreSQL SQL

Supabase Docker MLOps CI/CD

PROGRAMMING

Python TypeScript JavaScript

React NestJS Next.js HTML

CSS FastAPI Pydantic pandas

NumPy REST APIs SDK Design

Git

VISUALIZATION

Power BI Plotly Dash

Matplotlib ECharts Tableau

QUANT TRADING

QuantConnect IBKR API

Alpaca API

DATA

LSEG SEC Filings

Broker Research

Earnings Transcripts Third Bridge

EDUCATION

McGill University
B.Com Finance & Business Analytics
Desautels · 2019–2022
Montreal, QC

Champlain College
DEC Computer Science & Mathematics
2017–2019 · Honor Roll
Montreal, QC

CERTIFICATIONS

Azure AI Fundamentals
Microsoft · Sep 2023

LANGUAGES

English · Native

French · Native

Romanian · Native

Spanish · Basic

Italian · Basic